Physics of the Earth and Planetary Interiors, 16 (1978) 379–392
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PALAEOMAGNETISM AND K/Ar RESULTS FROM THE DUNCANSBY VOLCANIC NECK, NE SCOTLAND: SUPERIMPOSED MAGNETIZATIONS, AGE OF IGNEOUS ACTIVITY, AND TECTONIC IMPLICATIONS

K.M. STORETVEDT *, C.M. CARMICHAEL, A. HAYATSU and H.C. PALMER

Department of Geophysics, University of Western Ontario, London, Ont. N6A 5B7 (Canada)

(Received June 8, 1977; revised and accepted September 9, 1977)

Storetvedt, K.M., Carmichael, C.M., Hayatsu, A. and Palmer, H.C., 1978. Palaeomagnetism and K/Ar results from the Duncansby volcanic neck, NE Scotland: superimposed magnetizations, age of igneous activity, and tectonic implications. Phys. Earth Planet. Inter., 16: 379–392.

The Duncansby volcanic neck, intruding the Middle Devonian red beds of north Caithness, Scotland, has revealed two significantly different axes of magnetization, yielding pole positions at 149°E, 24°N and 126.5°E, 60°N, respectively. The first pole, which is interpreted as corresponding to the oldest magnetization, is in perfect agreement with Devonian polar estimates from west of the Great Glen Fault. It is tentatively suggested therefore that the Duncansby neck correlates with the Late Devonian volcanism in the nearby Orkney Islands though palaeomagnetism allows an upper age estimate of around Middle Carboniferous. The data support an earlier proposition of there being a palaeomagnetic discordance across the Great Glen Fault that can be interpreted in terms of a large-scale late- or post-Devonian transcurrent movement along this fracture zone. The original (? Late Devonian) magnetization has been nearly completely erased by the second phase of magnetization which, according to its pole position, most likely dates from about the Middle Jurassic. The latter magnetization is thought to be a consequence of burial, the coastal districts of Caithness having participated in the general subsidence of the North Sea area in late Palaeozoic and Mesozoic times. The burial magnetization, involving VPTRM and or TCRM processes, is considered to have been "frozen-in" as a result of uplift in connection with the well-documented mid-Jurassic tectonic phase that affected the northern North Sea basin, including the adjacent Moray Firth area. K/Ar analyses of the Duncansby intrusion give apparent ages ranging from 258 to 239 Ma. These dates, which lie between the two geological events inferred from palaeomagnetism, are not seen as true rock ages but rather as the result of a partial Ar loss during burial reheating.

1. Introduction

At Duncansby Ness on the northeastern tip of the Scottish mainland a small volcanic vent of ultramafic composition, exposed as rocky reefs between tide marks, is seen to penetrate the upper Middle Devonian John O'Groats Sandstone (Fig. 1). Petrographic descriptions have been given by Flett (1914) and Chapman (1975a). The rock, which to a major extent consists of agglomerate/tuff, has an enormous number of sandstone inclusions that range from small pieces to large blocks. Other inclusions include xenoliths of coarse-grained spinel–clinopyroxenite which have been studied by Chapman (1975b). The intrusion was originally considered as a branch of the volcanic activity that occurred in the southern part of the Orkney Islands in Late Devonian (Geikie, 1878) but recently two K/Ar dates (Chapman, 1975a; MacIntyre and Chapman, in prep.) have suggested an early Permian time for the event. In the present paper we report on the palaeomagnetic record of the Duncansby neck as well as providing further K/Ar results.

* Now at University of Bergen, Bergen, Norway.

2. K/Ar ages

Pieces of five of the block samples used for the palaeomagnetic study were dated by the K/Ar meth-

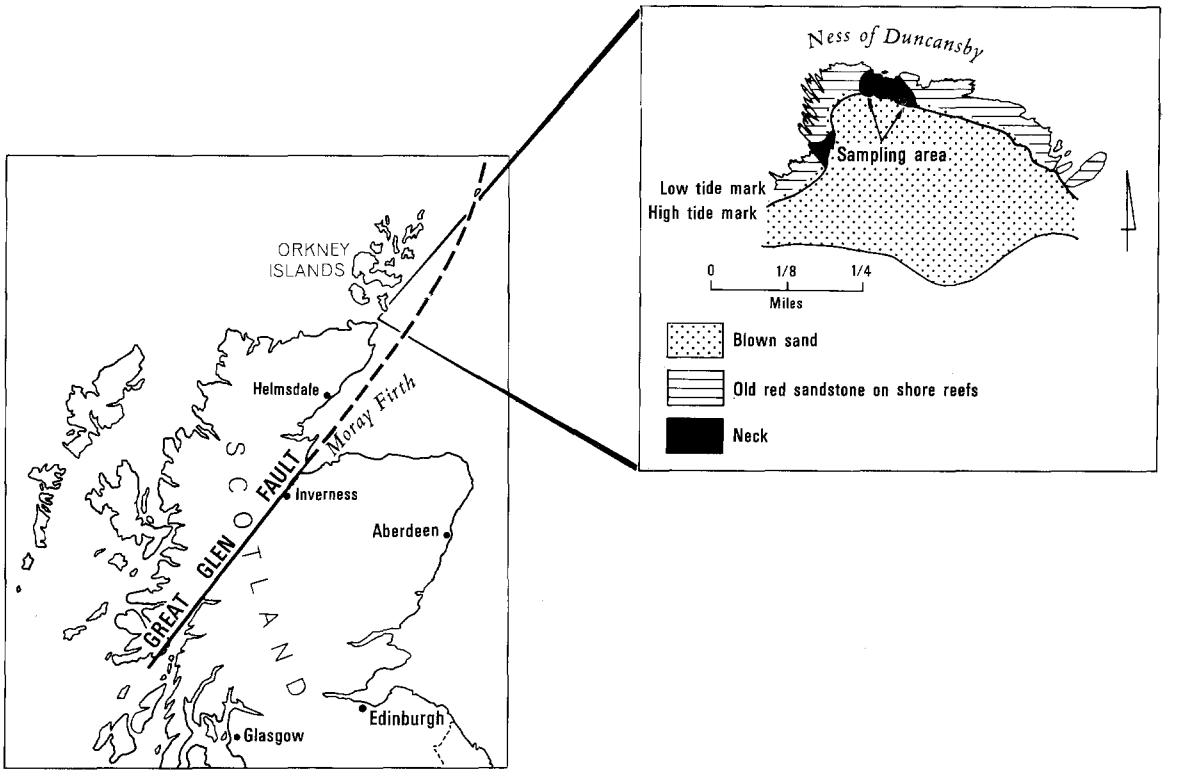


Fig. 1. Geographical and geological setting of the Duncansby volcanic neck. The geological sketch map is a simplified version after Flett (1914).

TABLE I

K/Ar analysis of the Duncansby volcanic neck

Sample	K (%)	$^{40}Ar_{rad}$ (10^{-5} cm^3/g)	$^{40}Ar_{rad}$/total (%)	Age (Ma)
JO–56	1.35 1.37 } 1.36 1.35	1.386	78.7	239
JO–58	0.786 0.783 } 0.785	0.801 0.810	78.4 74.2	241
JO–59	0.952 0.957 } 0.954	1.008	81.4	248
JO–61A	0.968 0.964 } 0.966	1.028	76.0	249
JO–61B	0.972 0.961 } 0.966	1.045	76.6	253
JO–65	0.938 0.929 } 0.933	1.034	63.1	258

$\lambda_e = 0.585 \cdot 10^{-10}$ a^{-1}; $\lambda_\beta = 4.72 \cdot 10^{-10}$ a^{-1}; $^{40}K/K = 1.22 \cdot 10^{-4}$.

od. Radiogenic Ar was determined by the isotope dilution method using an AEI MS-10 mass spectrometer. K was measured by an Instrumentation Laboratory flame photometer model 143 with Li internal standard and Na buffer. Core samples weighing about 15 g each were crushed to pass 20 mesh. Sandstone inclusions were hand-picked and discarded. The samples were then split into two portions, one for the Ar and one for the K analysis. The portion to be used for K analysis was further crushed to pass 100 mesh.

Results (Table I) from the duplicate analysis on sample J058 and of two samples (J061-A and J061-B) from the same block indicate that the analytical error is less than 2%. These ages are in agreement with the two given by Chapman (1975a) except that there is a spread toward younger ages. Our evaluation of the significance of the K/Ar dates is given below in Section 5.

3. Petrology, opaque mineralogy and magnetic properties

Polished thin sections, made from the palaeomagnetic cores of five of the samples, were examined in transmitted and reflected light. All of these sections are of xenolith-bearing ultramafic tuffs. Olivines, now pseudomorphed by serpentine and serpentine–calcite aggregates, together with clinopyroxenes, occur as microphenocrysts which are set in a fine matrix of clinopyroxene, oxides and a clear isotropic phase. The clinopyroxenes are sector zoned and display anomalous interference colors. Minor primary minerals seen in some of the sections are reddish-brown spinel and biotite. Plagioclase is restricted to secondary amygdule fillings where it is commonly accompanied by a colorless isotropic mineral having a refractive index of 1.53. The xenoliths include clinopyroxene–serpentine aggregates, feldspathic sandstones and calcareous shale chips.

Most of the opaque minerals consist of small grains of magnetite in the fine-grained silicate matrix. In addition there are small quantities of pyrite and pyrrhotite, and very minor amounts of poorly crystallized haematite in the microphenocrysts. No ilmenite was observed either as separate grains or as exsolution lamellae in the magnetite.

Euhedral grains of Cr-spinel, ranging up to 100 μm across, are common both within the matrix and partly or wholly imbedded within the olivine microphenocrysts. These Cr-spinel grains have rims of magnetite where they touch the matrix but not where they touch the serpentinized olivine (Plate I). More than half the serpentinized olivine and orthopyroxene microphenocrysts show no separated Fe oxides. The remainder have reddened patches when viewed in transmitted light, and some haematite fills cracks and forms partial rims. No magnetite was observed in the microphenocrysts either internally or as rims.

The abundant magnetite grains in the matrix are anhedral in shape and range from about 30 μm in diameter to less than 1 μm. They are all single phase in the sense that there is no exsolution of ilmenite or ulvöspinel observable when viewed at a magnification of 1,000 under oil immersion. In some samples a number of grains have patches of maghemite on their edges (Plate I). After the repeated heating of the thermal demagnetization procedure it is found that some of the silicates are reddened and in these areas the magnetite grains are partly or completely converted to haematite, but most of the sample, including the magnetite grains, are unchanged. Since none of the unchanged magnetite grains in the heated samples was observed to contain maghemite it is assumed that all the maghemite has been converted to haematite.

Curie balance curves of magnetization vs. temperature measured in a field of 8 kG (0.8 T) for representative samples are shown in Fig. 2. These curves are very similar, over the higher temperature range, to the thermal demagnetization curves (cf. Figs. 3–5) indicating that the blocking temperature spectrum is similar to the Curie temperature spectrum in this range. All samples have curves that are reversible above a temperature of 400°C but below about 350°C the cooling curves fall below the heating curves, suggesting the conversion of a magnetic phase to a non-magnetic one. This effect is interpreted as being due to the maghemite being converted to haematite; the latter mineral is only very weakly magnetic in a field of 0.8 T. This is supported by the relative amounts of maghemite observed in the polished thin sections; none was observed in J064, samples with curves similar to J062 had most and those with curves similar to J058 and J056 had lesser amounts. A large part of the magnetic component has a Curie point between about 500 and 550°C suggesting a

PLATE I

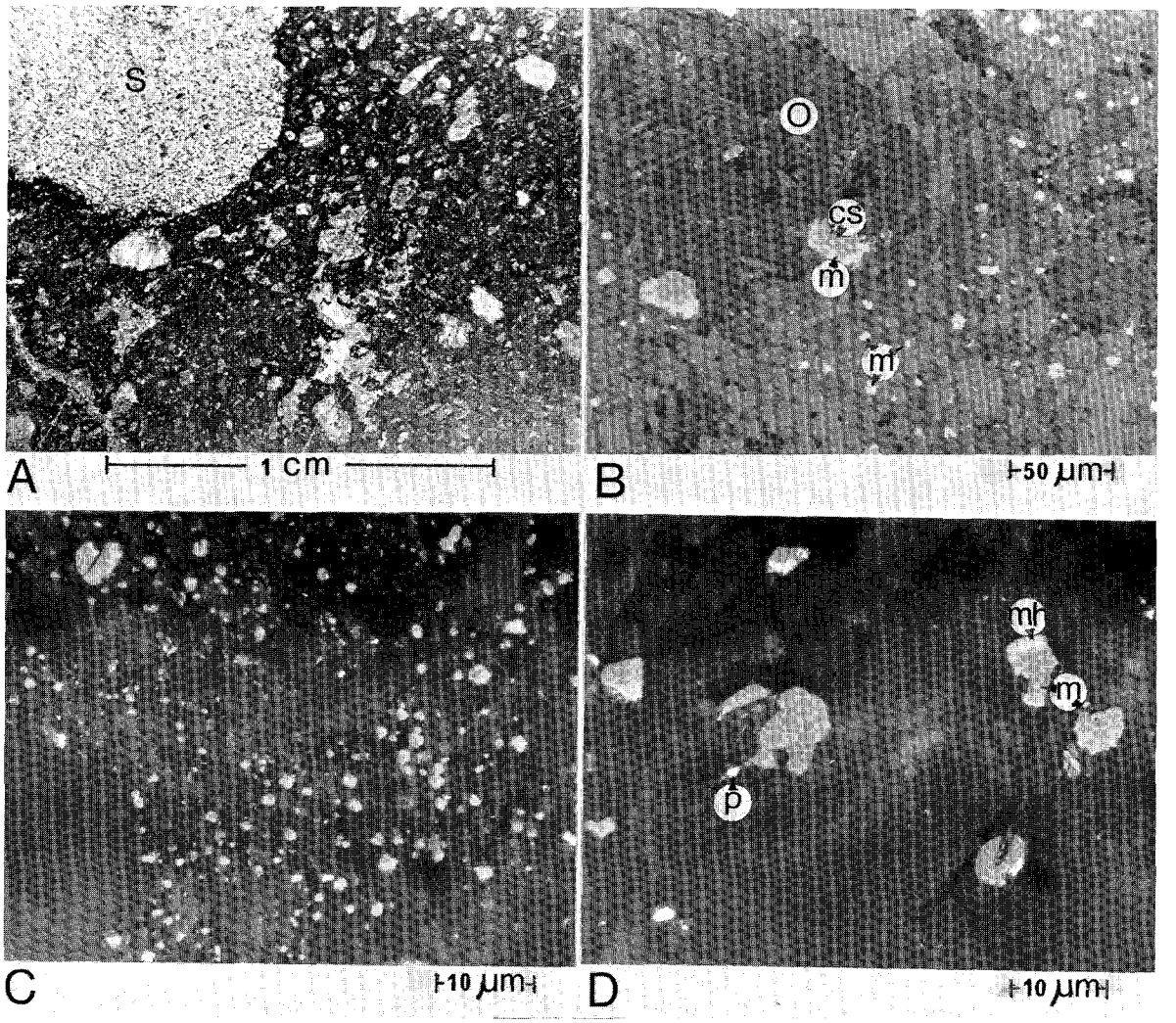


PLATE I

Photomicrographs of polished thin sections of samples from the Duncansby neck.

A. A low-magnetification photo of J056 by transmitted light; *S* is a sandstone inclusion; smaller light grains are microphenocrysts; the fine-grained silicate matrix containing abundant magnetite grains shows as the dark area.

B. An intermediate scale magnification by reflected light of J060 showing a serpentinized olivine microphenocryst (*O*); partly imbedded in the olivine are two Cr-spinel grains (*cs*), which have magnetite rims (*m*) where they touch the silicate matrix. In the matrix are numerous magnetite grains (*m*). None of the magnetite has been produced by serpentinization of the olivine.

C and D. Large-scale magnification by reflected light under oil immersion; C is a part of J064 showing very small magnetite grains as white dots; D shows larger magnetite grains (*m*) in J061, some with maghemite patches (*mh*) on their edges. A pyrrhotite grain is shown by (*P*). The lines on the larger magnetite grains are scratches not exsolution lamellae. The grains analysed by electron microprobe (see Table II) were similar but larger than those shown here.

magnetite with 5–15% ulvöspinel in solid solution. The shape of the cooling curves, particularly those similar to J062, suggests that there are also components present with lower Curie points and hence larger proportions of ulvöspinel.

The X-ray diffraction patterns of the separated

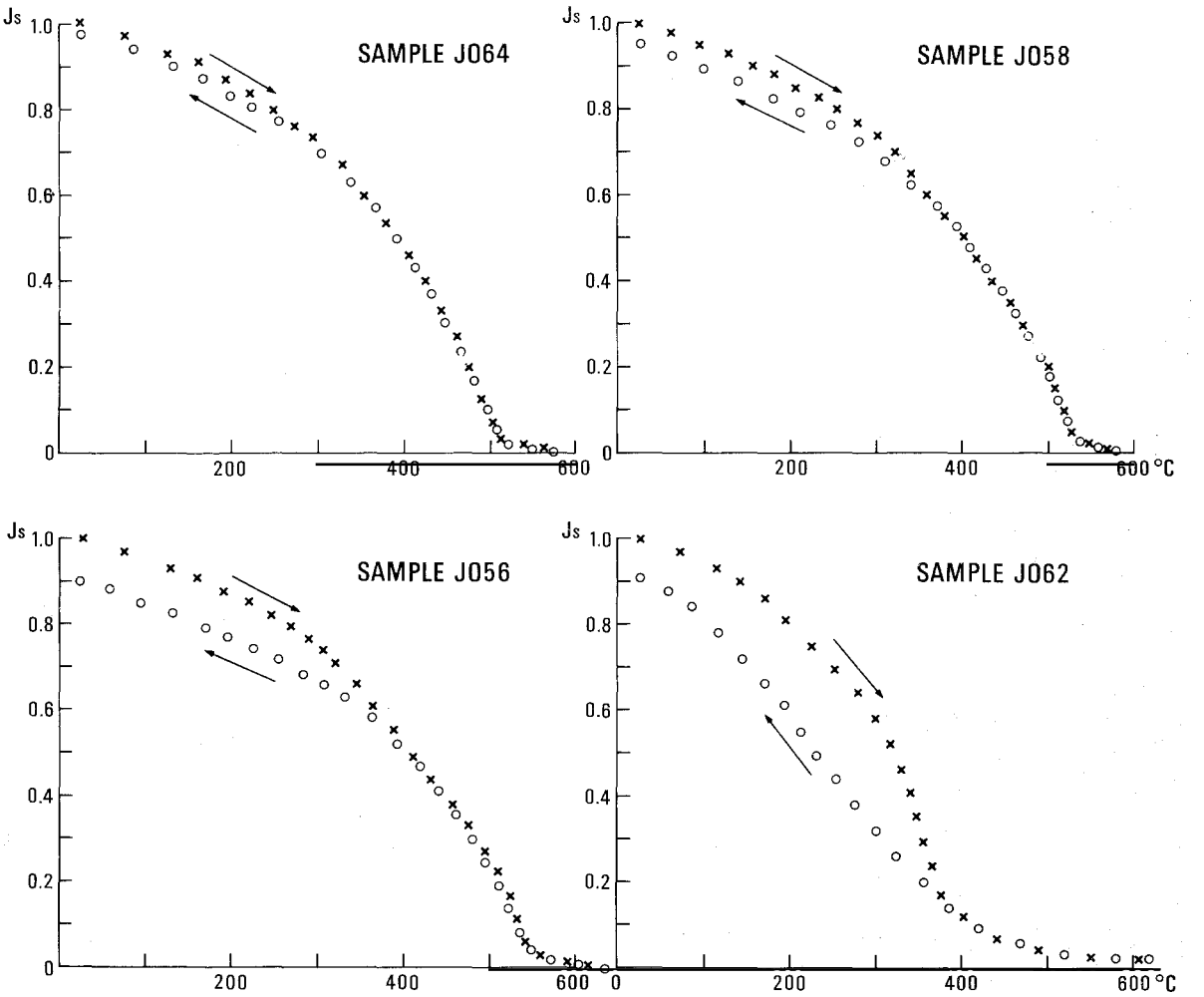


Fig. 2. Examples of saturation magnetization vs. temperature. Crosses are data obtained on heating while circles are results attained during the subsequent cooling.

magnetic fraction have faint and diffuse lines. Microscopic examination of the powder particles, after they were cast in plastic and polished, showed that they contained a preponderance of the largest of the magnetite grains but, due to the small size of even the largest ones, the particles were mostly composed of silicate. The best pattern gives a unit cell of 8.49 ± 0.02 Å. Assuming the element that is stretching the lattice is Ti, this corresponds to a magnetite–ulvöspinel solid solution having a Curie temperature of about 100°C. Since the X-ray pattern would be due mostly to the larger magnetite grains it suggests that there is a range of Ti content in the magnetite with less Ti in the smaller grains to account for the higher Curie points shown in Fig. 2.

Electron microprobe studies of the magnetite grains tended to confirm that the larger magnetites have considerable Ti. The beam width is too wide to analyse the smaller grains but analyses of a number of large to intermediate grains are given in Table II. Analyses are also given for two Cr-spinels, the magnetite rim of a Cr-spinel, a serpentinized olivine microphenocryst and the fine-grained matrix in which the magnetite grains occur. Ni and Zn in the magnetites were below detection level. Ca and Na were not analysed and may account for most of the missing constit-

TABLE II

Electron microprobe analyses of minerals in polished thin sections of samples JO64 and JO63 of the Duncansby neck. Analyses of the elements are given as oxides in percent

Microscopic identification	Mg	Al	Si	Cr	Mn	Ti	Fe	Total
Magnetite grain ~25 μm	2.69	7.75	0.50	4.04	0.89	13.57	66.85	96.28
Magnetite grain ~25 μm	2.14	6.51	0.48	0.00	1.07	17.44	65.82	93.46
Magnetite grain ~20 μm	2.27	7.07	3.13	0.64	0.96	15.35	60.04	89.46
Cr-spinel	7.80	14.05	0.13	50.17	0.91	1.42	24.66	99.13
Cr-spinel	18.18	38.57	0.26	23.61	0.26	0.59	15.25	96.73
Magnetite rim	3.03	7.78	3.27	3.91	0.83	14.84	62.38	96.03
Serpentinized olivine	45.33	0.00	39.83	0.00	0.04	0.00	12.48	97.68
Fine silicate matrix	2.92	20.90	45.39	0.00	0.00	0.58	4.24	74.03

Mineral grains similar to those analysed are shown in Plate I.

uents in the fine-grained matrix analysis. In addition to Ti, some of the magnetites contain appreciable Cr while others do not, but all have roughly 2% Mg and 7% Al. If the effect of these elements is ignored the Ti content is equivalent to 50–60% ulvöspinel which should have Curie points in the range 250–200°C.

Based on magnetic, microscopic, X-ray and microprobe evidence it is concluded that the magnetite grains have a Ti content that decreases with grain size. The hard remanent magnetization with Curie temperatures and blocking temperatures above about 500°C is interpreted as being due to the very fine magnetite grains which are also lowest in Ti, while the constituents with lower Curie points and blocking temperatures are inferred to be the magnetite grains of larger size and greater Ti content.

4. Palaeomagnetic measurements

The enormous number of large inclusions and the limited exposure makes the Duncansby neck difficult to sample. Ten oriented blocks of relatively fine-grained material were collected along the inner part of the main exposure (see Fig. 1).

The palaeomagnetic analysis was initiated by alternating field (AF) treatment to at least two specimens from each block sample. A soft random component of magnetization was easily removed by AF treatment in fields of 100 Oe (10 mT) or less, after which the remanence grouped extremely well around a mean direction of about $D \sim 208°, I \sim -55°$. This group, which will be called group A, remained stationary on subsequent AF treatment up to some hundreds of oersteds. A few specimens were successfully demagnetized in fields of 900 or 1000 Oe (90 or 100 mT) but in general an erratic behaviour in both intensity and direction, seen for example by a non-repeatability of results when re-demagnetizing specimens in the same field, was beginning to become significant at about 600 Oe (60 mT). For this reason AF treatment was discontinued. At this level of demagnetization the remanence intensity had dropped to 5–10% of the initial NRM value which was mostly in the range of $(9–20) \cdot 10^{-4}$ e.m.u./cm^3 (0.9–2 A/m). At this stage of demagnetization treatment it seemed as if the Duncansby neck had a very simple remanence buildup with only one component of stable magnetization. However, it was decided to try thermal demagnetization on top of the AF treatment.

Thermal demagnetization was carried out in an ambient field of the order of 1 nT, in a furnace heated by a non-inductively wound Pt heating coil and magnetically shielded by six nested cylindrical mu metal shields. Provision is made for slow rotation of samples within the furnace, to reduce thermal gradients and to help randomize magnetizations picked up on cooling, but it was found that this procedure was not necessary and was not used on most runs. To increase the spacing between the four specimens in the furnace, each AF-treated specimen (in the form of a 2 cm cylinder) was cut perpendicular to the axis to form two smaller specimens identified as x (top) and y (bottom). With one exception,

as noted below, the remanence parameters of the small specimens were the same as those of the larger specimen before it was cut.

The two-stage analysis, AF followed by thermal, gave very unexpected results. There appeared to be a tendency for the remanence vector to swing into a southerly direction with shallow downward inclination. However, in the initial phase of the thermal study this latter component often became obliterated due to scatter and by problems in defining the remanence vector after each heating step. It is worthwhile recording some of the experimental problems encountered in reducing this noise level. The first problem resulted from increasing the cooling rate by blowing compressed air into the furnace. This seems to have left the specimens with non-uniformly distributed regions having different magnetic relaxation states. Upon transfer of the specimens from the furnace to the magnetometer they could acquire a large nonhomogeneous viscous magnetization. These viscous components could sometimes be removed by storage in zero field for half a day. However, by allowing specimens to cool slowly in the furnace over periods of several hours they were left in a state in which viscous components were small. A second source of noise was a shock-induced magnetic moment that could be as much as ten times as large as the actual remaining NRM. Bumping the sample against a hard surface or dropping it could produce this spurious magnetization but very careful handling could prevent its production. Both of these spurious magnetizations must reside in a very soft magnetic constituent resulting from the heat treatment, but it is not obvious what this constituent can be. It may be the maghemite phase as it is being ordered before assuming the haematite structure or it may be some of the largest magnetite grains that are having their internal strains annealed by heating.

After precautions had been taken against these problems a shallow reversed magnetization was found in all collected blocks, though not in all specimens. In general, this component is only seen below an intensity level of about $5 \cdot 10^{-5}$ e.m.u./cm^3 ($5 \cdot 10^{-2}$ A/m). Representative examples of this type are shown in Fig. 3. In some cases the terminal direction has been defined by two or more heating steps while in other cases the remanence vector moved into the shallow reversed directional group at the last step of demagnetization, i.e. just before the magnetization became too weak for reliable measurement, as indicated by the magnetic moment changing continuously on repeated measurement. The shallow inclined reversed group is hereafter referred to as group B.

A number of specimens suggest that the magnetization buildup in the Duncansby neck is more complex than the two-component structure shown by Fig. 3. Specimen J063-D1 (Fig. 4) provides an interesting example in this respect. After AF treatment to 500 Oe (50 mT) the apparently stable end-point of this specimen was slightly more southerly and shallow than that characteristic for the A-group. Also, the remanence measurements showed a larger inhomogeneity of magnetization than normally encountered on AF treatment of these rocks. After slicing this specimen, the direction of magnetization for the y-specimen came into perfect agreement with that of the A-group, while the x-specimen attained a fairly steep positive (downward) and easterly direction of magnetization. The resultant magnetization of the two smaller specimens are in harmony with the magnetization of the larger specimen before cutting. It was found that both specimens responded well to AF treatment up to 1000 Oe (100 mT): the y-specimen remained stationary in the A-group while the x-specimen defined a northerly loop (cf. Fig. 4) before turning south between 900 and 1000 Oe (90 and 100 mT). The subsequent thermal demagnetization caused the direction of the x-specimen to move immediately into the B-group. Vector subtraction of the B-component, the direction of which is defined by the overall mean of the 220, 300 and 560°C steps and the intensity at the 220°C step, from the 500–1000 Oe (50–100 mT) vectors reveals that the most stable remanence of this specimen constitutes the two antiparallel components of the shallow inclined magnetization axis. In view of the problems of estimating a reliable palaeomagnetic subcomponent through vector subtraction, since the intensity of the B-group component may already have experienced a certain reduction at the 220°C heating step, the normal sub-component seems surprisingly well defined. Thermal demagnetization of the y-specimen suggests the same two-component buildup of the most stable remanence: the directional data define a great circle in the plane of the x-specimen components but the termi-

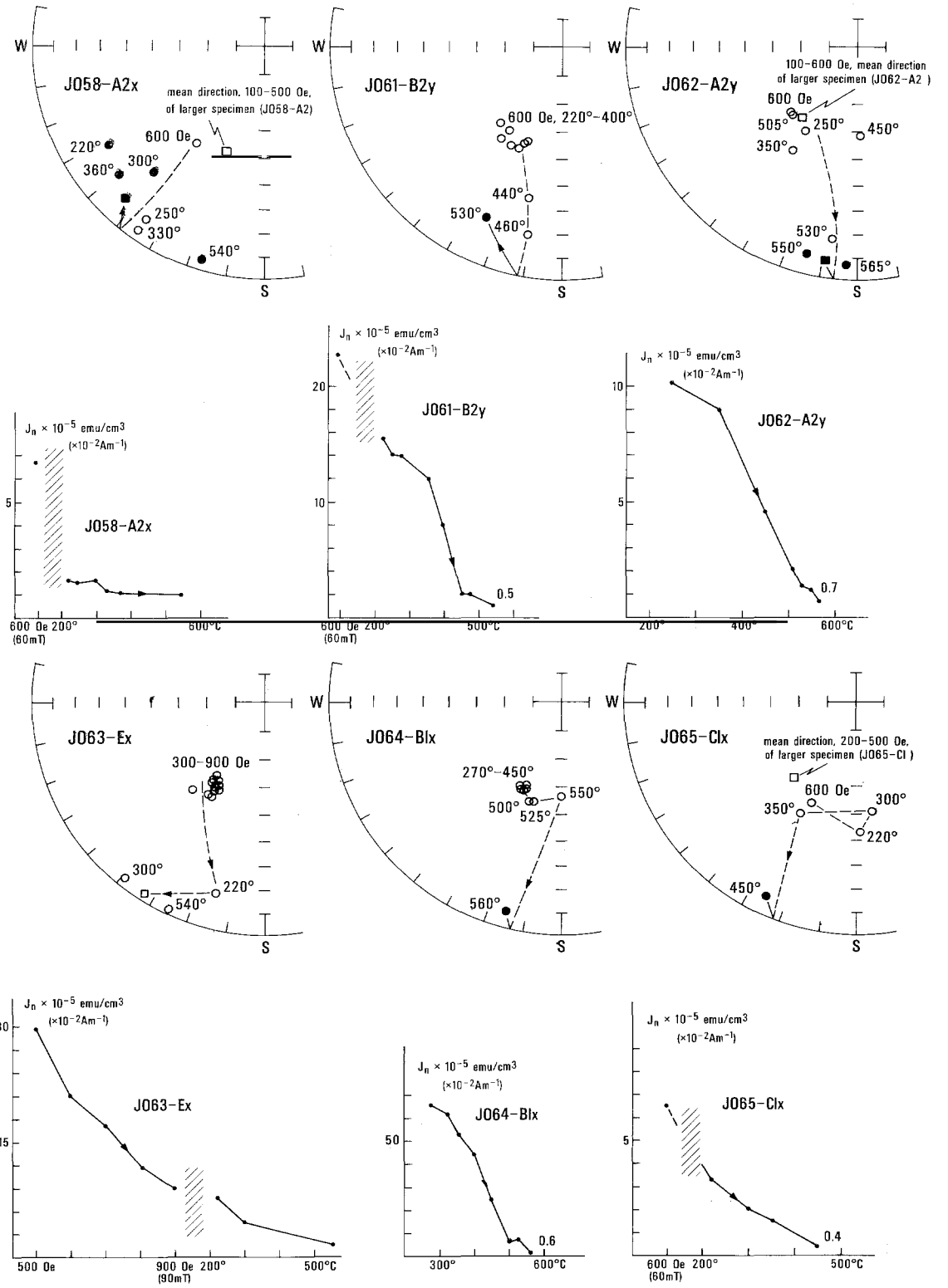


Fig. 3. Representative examples of direction and intensity pattern vs. progressive thermal demagnetization. All thermal results are obtained after previous alternating field treatment up to 600 or 900 Oe. Open (closed) symbols are upward (downward) pointing magnetization vectors. Where not otherwise stated squares denote the mean high-temperature direction (shallow reversed) for individual specimens. Such mean directions form group B1 of Table III. Projection is equal area.

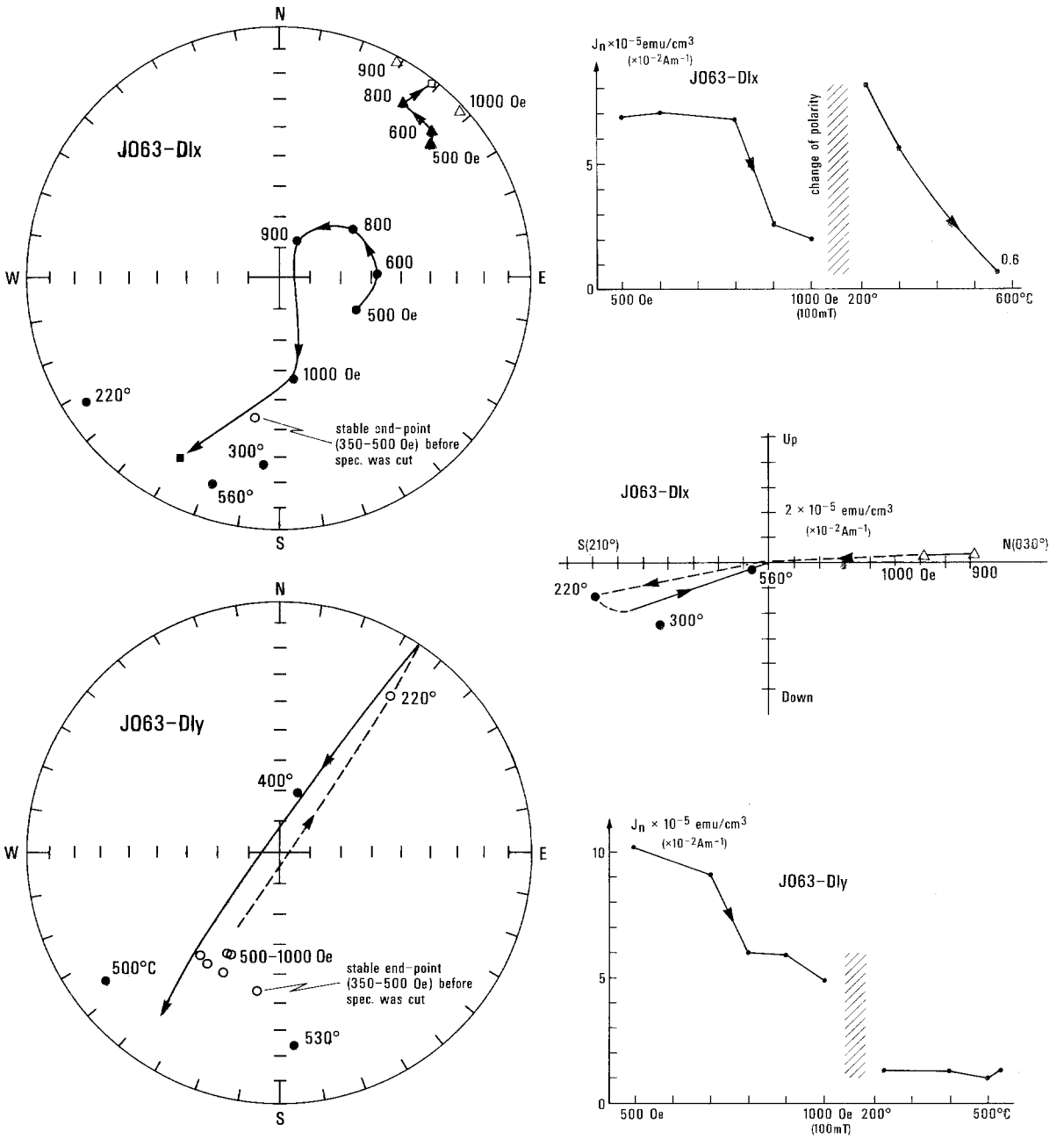


Fig. 4. The buildup of the most stable magnetization in specimen J063-D1 illustrated by slicing the specimen into two smaller, equally sized, cylinders (x and y) and subsequently applying a two-stage demagnetization treatment: alternating field (up to 1000 Oe) + thermal. Triangles represent magnetization vectors obtained by vector subtraction (J063-D1x). In addition to the conventional direction and intensity diagrams the variation of the two most stable sub-components of specimen J063-D1x is shown in a vertical plane striking NE–SW. Further details in text. Diagram conventions as for Fig. 3.

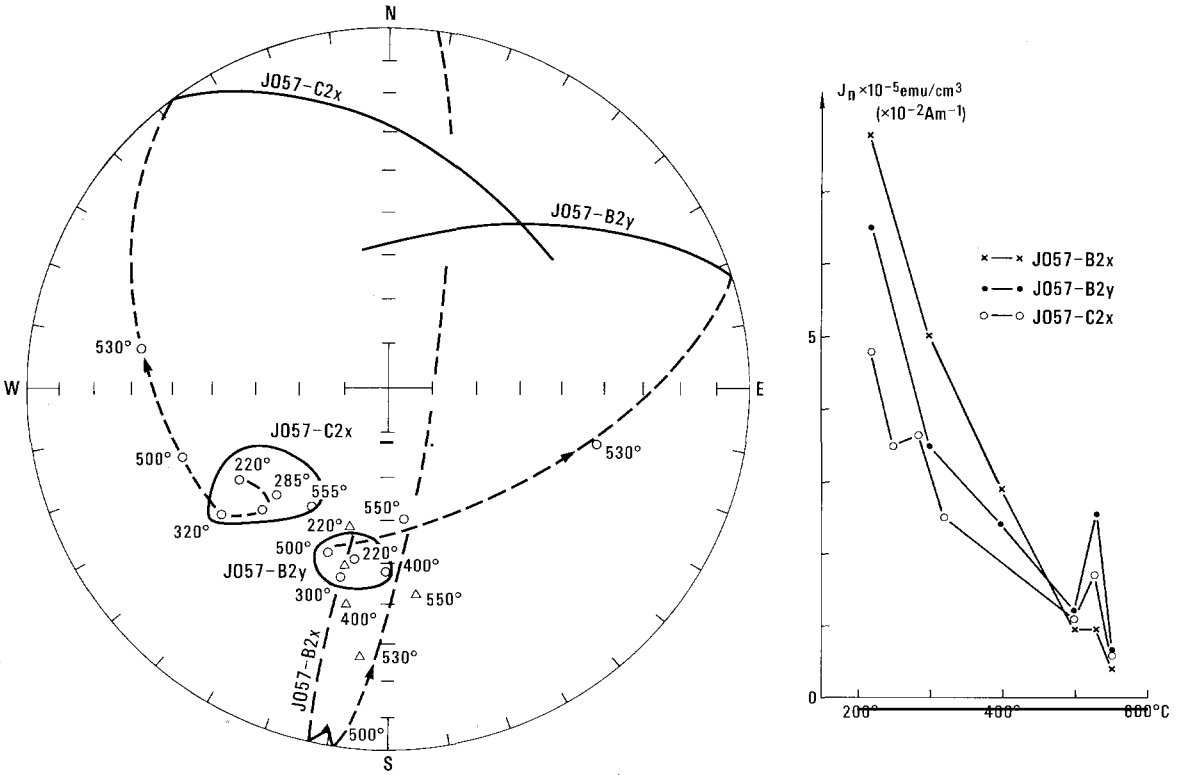


Fig. 5. Thermal demagnetization results from sample J057 which exhibit a more varied behaviour than the rest of the collection. Note that for specimens B2y and C2x the remanence vectors return to the initial groupings after heating above 530°C. Results for specimen B2x are shown by triangles in order to distinguish them from those of specimen B2y. Consult text for further information. Diagram conventions as for Fig. 3.

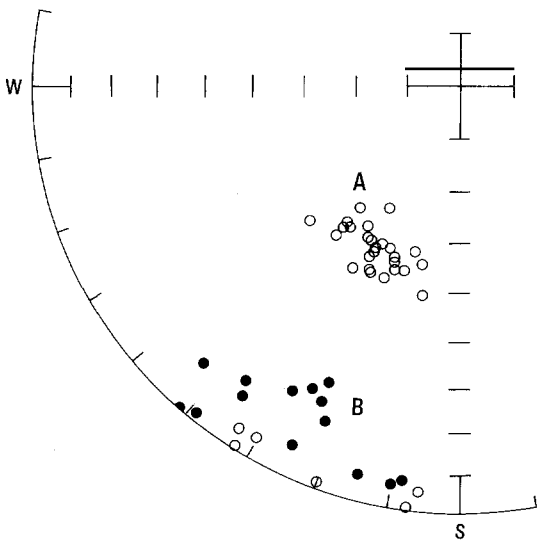


Fig. 6. Distribution of the two important magnetization components in the Duncansby neck. The shallow inclined group corresponds to group B2 in Table III. General information as for Fig. 3.

nal direction (reversed) is less well defined.

Thermal demagnetization data of specimens from sample J057 tend to indicate that the antiparallel component of the A-group is also present. Fig. 5 demonstrates the directional behaviour of specimens from this sample as a function of increasing temperature. Two of these specimens J057-B2y and J057-C2x, suggest great-circle paths which tend to have a normally directed intersection point approximately antiparallel to the A-group. Also, the A-group itself (Fig. 6) has a slight elongated distribution, the direction of which supports the idea of there being a certain normal component with moderately steep downward inclination. For specimens J057-C2x and J057-B2y the inferred normal sub-component has an "intermediately" high thermal stability in that the high-temperature direction return to the A-group above 530°C. Specimen J057-B2x makes a short visit to the B-group at 500°C before heading north (prob-

TABLE III

Palaeomagnetic data from Duncansby volcanic neck

Group	N	D_m (°)	I_m (°)	κ	α_{95} (°)	Pole		
						long. (°E)	lat. (°N)	
A	26	207.9	−54.6	155	2.3	126.5	60.3	
B1	10	208.4	+09.0	25	9.8	146.1	23.0	including only terminal specimen directions defined by two or more demagnetization steps
B2	20	205.3	+08.0	28	6.3	149.1	24.3	including also specimens for which the B-component appeared at the last step of demagnetization

N = number of specimens; κ and α_{95} are parameters according to Fisher (1953).
D_m = mean declination; I_m = mean inclination.

ably towards the antiparallel component of the B-direction).

In summary, the Duncansby volcanic neck seems to be the carrier of two distinctly different axes of magnetization. Fig. 6 illustrates the distribution of the two major components, groups A and B. Group A carries the dominant part of the remanence intensity. Group B is in general associated with the last 1–5% of the total magnetic moment. Without taking the necessary experimental precautions this latter magnetization component might easily have gone unnoticed. The B-group is slightly more scattered than the A-group but this is understandable taking into account the problems with magnetic instability and scatter at these low intensities. With reference to Table III, different acceptance criteria for the B-group have only negligible influence on the overall palaeomagnetic parameters.

5. Discussion and conclusions

The palaeomagnetic results outlined above suggest that the Duncansby neck has experienced at least two magnetization "pulses" and the large inclination difference between the A- and B-group (65°) signifies that there is a long time span between them. Fig. 7 shows the estimated Duncansby poles (A and B) in conjunction with the suggested European polar path for the time interval: Ordovician to Cretaceous. Pole *B*, which corresponds to the oldest of the two magnetizations, plots in extremely good agreement with available Devonian data from west of the Great Glen Fault. As the Duncansby Neck intrudes Middle Devonian strata the B-component must be Late Devonian or younger. The upper age limit of the component is probably about Middle Carboniferous as the subsequent palaeomagnetic axis shift, preceding the onset of the Kiaman geomagnetic interval, occurred at around this time (McElhinny, 1973). However, there was volcanism in the nearby Orkney Islands (cf. Fig. 1) in Late Devonian time so at present it appears most reasonable to conclude that the Duncansby intrusion relates to this activity. Based on the extremely good match of pole B with other Old Red Sandstone poles from west of the Great Glen Fault the initial magnetization processes are likely to have persisted for a sufficient length of time to record the total variation of geomagnetic secular variation. Also, the observation of an antiparallel direction (Specimens J063-Dlx and y) to the B-group supports the suggestion that the magnetization processes were of somewhat longer duration. In any case, the Duncansby pole B has strengthened the proposition of Storetvedt (1974) of there being a palaeomagnetic discordance across the Great Glen Fault which can be explained in terms of a large-scale sinistral displacement along this fracture zone.

The pole A plots in a Mesozoic position on the European polar wander path. Unfortunately, many of the Mesozoic palaeomagnetic poles for Europe are based upon earlier studies when laboratory treatment

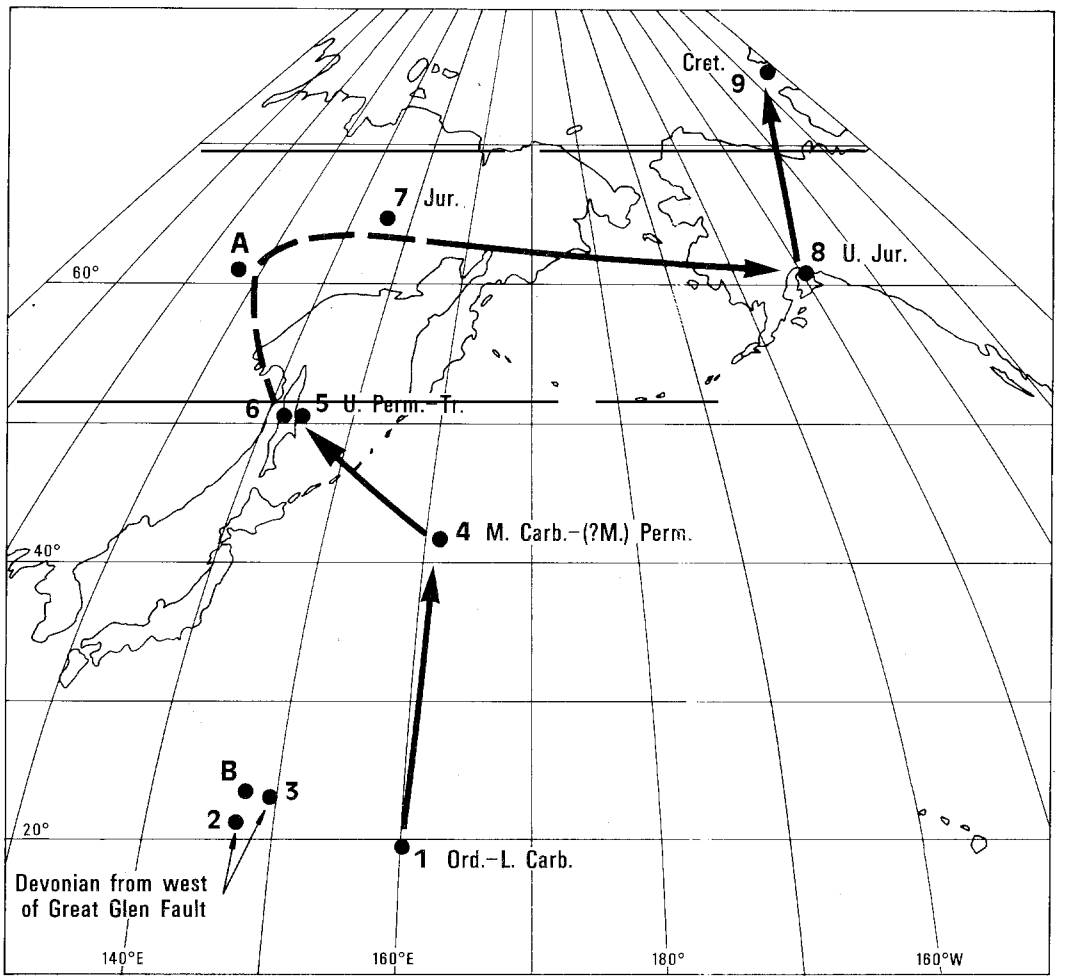


Fig. 7. The Duncansby poles A and B seen in conjunction with the suggested European polar wander path for part of Phanerozoic time. Poles *1*, *4*, *5* and *7* are mean poles from the following sources: *1* = Storetvedt (1967); *4*, *5* and *7* = Phillips and Forsyth (1972) (other sources give similar results). Individual formation poles as follows: *2* = Waage and Storetvedt (1973); *3* = Storetvedt and Petersen (1972); *6* = Zijderveld (1975); *8* = Briseid and Halvorsen (1974); *9* = Sandal and Halvorsen (1973).

of samples was minimal. However, recent results from southern France (Zijderveld, 1975) seem to verify the earlier pole determinations for the Late Permian to Triassic. There is no modern palaeomagnetic study of the middle Mesozoic but recent data from late Jurassic–Cretaceous volcanics of Spitsbergen reveal a similar westward polar shift in the Jurassic of Europe as can be seen in the better defined Mesozoic polar-wander curve for North America. There are some reasons to believe therefore that despite great uncertainties in the available Jurassic palaeomagnetic data for Europe the overall pole position may not be essentially wrong. With these limitations on the present polar pattern the Duncansby pole A plots in a position that is compatible with an age of this magnetization component between say Late Triassic to Middle Jurassic. It is pertinent to ask whether also pole A (like pole B) is actually situated west of the true polar track for continental Europe. Recent geophysical studies in adjacent offshore sedimentary basins related to the Great Glen Fault (Bacon and Chesher, 1975; Bott and Browitt, 1975) suggest that any transcurrent movement along this fracture zone in the Mesozoic and later must have been very small. Any

noticeable upsetting effect on the pole A for tectonic reasons is therefore considered unlikely.

The A-component is not only an important magnetization in the Duncansby intrusion but such a magnetization also strongly overprints the Devonian palaeomagnetism of the John O'Groats Sandstone (Storetvedt and Carmichael, in prep.) as well as giving some evidence of being present in the Old Red Sandstone deposits further south in Caithness (Waage and Storetvedt, 1973). This suggested widespread distribution of the A-component in the northeasternmost districts of Scotland may be better understood when seen in a regional geological context.

In addition to tectonic activity in the early Mesozoic, represented by the so called Hardegsen unconformity, the greater part of the North Sea region with adjoining land areas underwent subsidence during the late Palaeozoic and early Mesozoic (Kent, 1975). During the Jurassic, however, the North Sea region along with most of NW Europe was affected by more pronounced extensional forces causing both uplift and subsidence of fault blocks (Kent, 1975; Ziegler, 1975) as well as causing the main volcanic episode in the northern North Sea graben. In the Moray Firth basin (see Fig. 1) the quite extensive faulting was most active in pre-Middle Jurassic times, forming two depositional troughs separated by a horst-like uplift zone (Bacon and Chesher, 1975). Along the Great Glen Fault in the Moray Firth the Jurassic sequence has been down-faulted against the Old Red Sandstone and similar features are exposed along the Helmsdale Fault in the northeastern coastal districts of Sutherland. It seems reasonable to assume therefore that throughout late Palaeozoic and early Mesozoic at least, parts of NE Scotland belonged to the North Sea depositional basin, being subsequently subjected to uplift and erosion in Lower–Middle Jurassic time. A prolonged burial of the region, giving rise to a temperature elevation which could induce chemical alteration and would increase the rate at which viscous magnetization would be acquired, may explain why the original magnetization of the Duncansby neck has been nearly completely erased. We therefore consider the A-component to be the result of a secondary magnetization associated with burial, having been "frozen in" as a result of the lowering of the temperature upon uplift in Lower–Middle Jurassic time. In the present case one is dealing with a marginal area of the North Sea sedimentary basin and, despite the close proximity to the Great Glen Fault, along which extensive normal faulting may have taken place, it is difficult to envisage a very deep burial. The A-component is carried by constituents with blocking temperatures generally below about 500°C. Pullaiah et al. (1975) have calculated that it would require prolonged heating at temperatures up to 350°C to remagnetize pure magnetite with this blocking temperature. The magnetite in these samples is not pure, which may allow remagnetization at lower temperatures, for it seems unlikely that burial would be deep enough to reach 350°C. The process of remagnetization may have also been aided by chemical changes as well. There is some evidence for such processes in that some maghemite has been produced and the olivines and orthopyroxenes have been serpentinized. When all the evidence is considered however, we are led to believe that the A-component has been produced by processes in which thermal effects are more important than chemical ones.

While the palaeomagnetic evidence suggests intrusion during the Late Devonian and partial remagnetization during the Lower–Middle Jurassic the radiometric ages do not conform to either of these periods. The K/Ar ages reported in this paper and given in Table I confirm those reported by Chapman (1975a) though they show a 20 Ma spread toward younger values. It is contended therefore that the apparent Permian K/Ar ages estimated from the Duncansby volcanic vent do not represent the actual age of this intrusion.

We conclude that the Duncansby neck intruded the Middle Devonian John O'Groats Sandstone in Late Devonian time in an explosive event that incorporated large amounts of solid material in a fluid matrix. Cooling within the neck crystallized the fluid into a fine-grained matrix but proceeded at a rate that allowed the formation of small anhedral grains of magnetite having a wide range of blocking temperatures due to variation in grain size and Ti content. The original magnetization processes were slow enough to record the Late Devonian dipole field direction. Subsequent reheating during burial has remagnetized all but the highest blocking temperature constituents, which is thought to be represented by the finest and purest magnetite grains. During this burial some of the radiogenic Ar was lost from the

neck so that K/Ar dates give a range of ages that fall between the times of intrusion and uplift after burial. Cooling upon uplift during the Middle Jurassic preserved the average palaeomagnetic field direction at that time.

Acknowledgements

This research was supported by NRC grant number A1776 and by the Norwegian Research Council for Science and the Humanities through grant D.41.12-6. A grant from Kr. Birkelands Fond to cover field expenses is gratefully acknowledged. The electron probe analysis and the X-ray diffraction measurements were performed by R.L. Barnett and Yu-Chi Chang, respectively, in the Department of Geology. Leif Erik Otterå kindly assisted in the field.

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